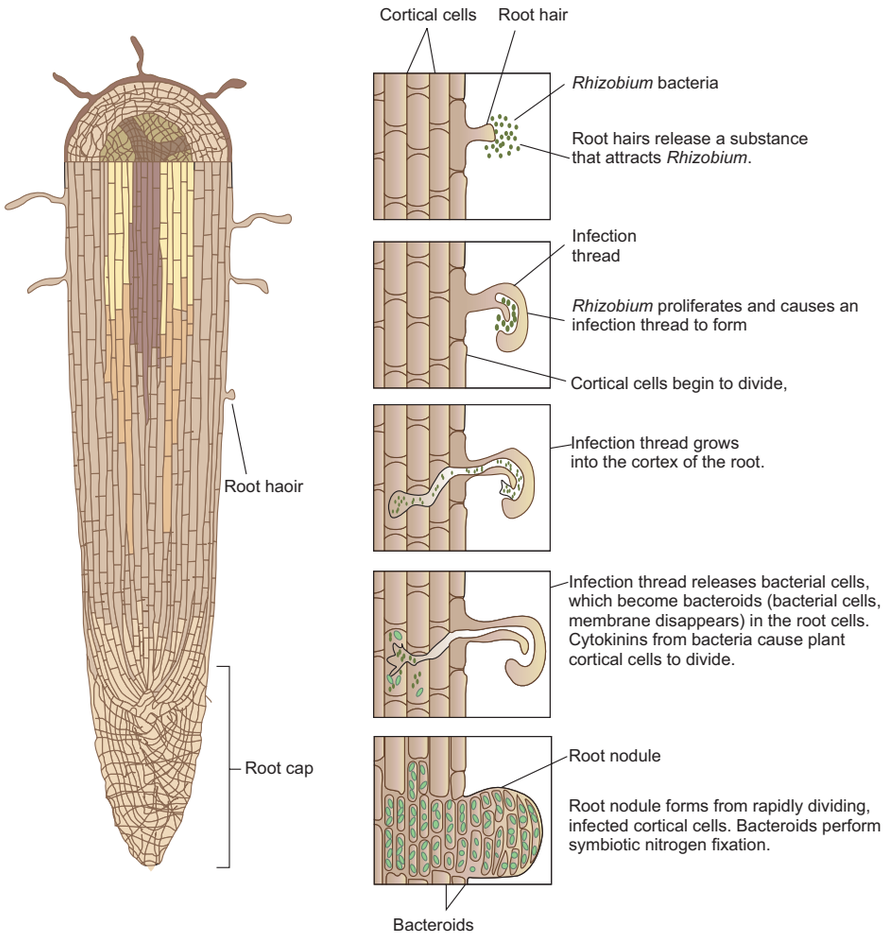


Figure 11.4: Showing stages of formation of root nodules on legume plant

Phosphate solubilizations

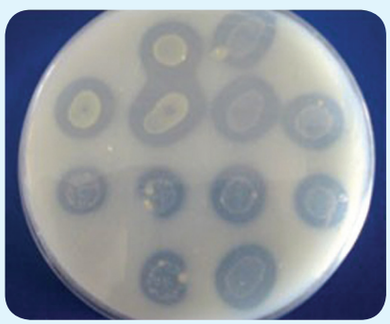
Most of phosphates occur in combination with Calcium, Iron, Magnesium and Aluminium (inorganic P) and thus are insoluble and unavailable to plants and micro organisms. Some micro organisms solubilize those insoluble phosphates by producing organic acids. Example: *Thiobacillus, Bacillus* thus enabling the plants to utilize it.

Phosphate assimilation

Plants and micro organisms can readily assimilate soluble forms of inorganic phosphates like $H_2PO_4^-$, HPO_4^{-2} and HPO_4^{-3} and incorporate them as organic forms of phosphates like ATP, nucleic acids.

Infobits

The ability of bacteria to solubilise phosphates can be tested in the laboratory by streaking the bacterial culture in Pikovaskaya agar which contains tricalcium phosphate. Positive cultures show clear halo around growth.



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